

WSPRadar.org reconstruction & comparison

SNR difference by time of day

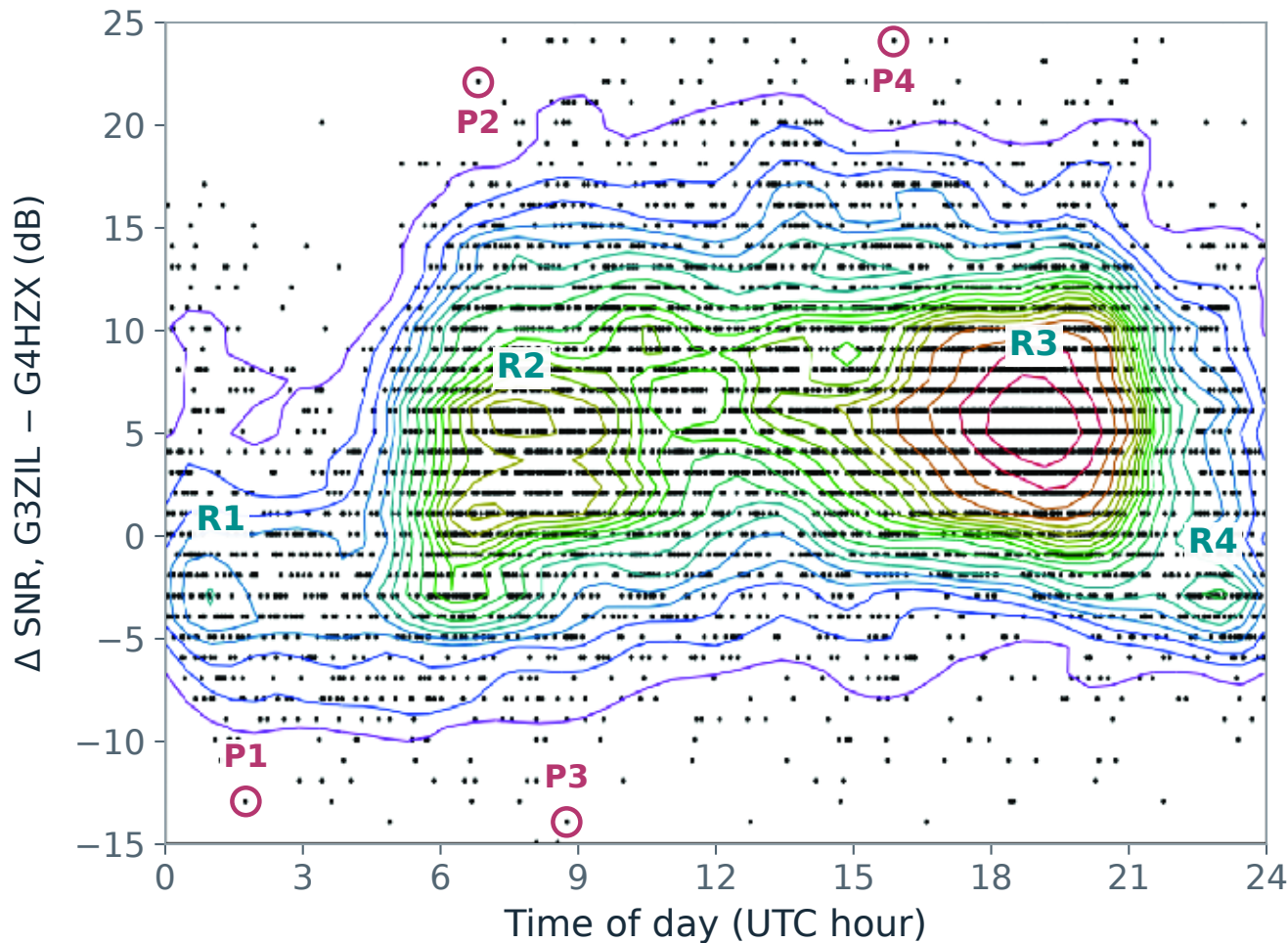
Referenced publication: Gwyn Griffiths (G3ZIL) and Nigel Squibb (G4HZX) (2017). *Improving HF Band SNR*.

Source figure: **Figure 6, p. 25** · Practical Wireless, October 2017, pp. 23-26

Demo: 5-7 April 2017 | 40 m | $\Delta\text{SNR} = \text{G3ZIL} - \text{G4HZX}$ | B/C: 6,459 paired observations

A Original image from publication

Original scatter and contours · linear dB



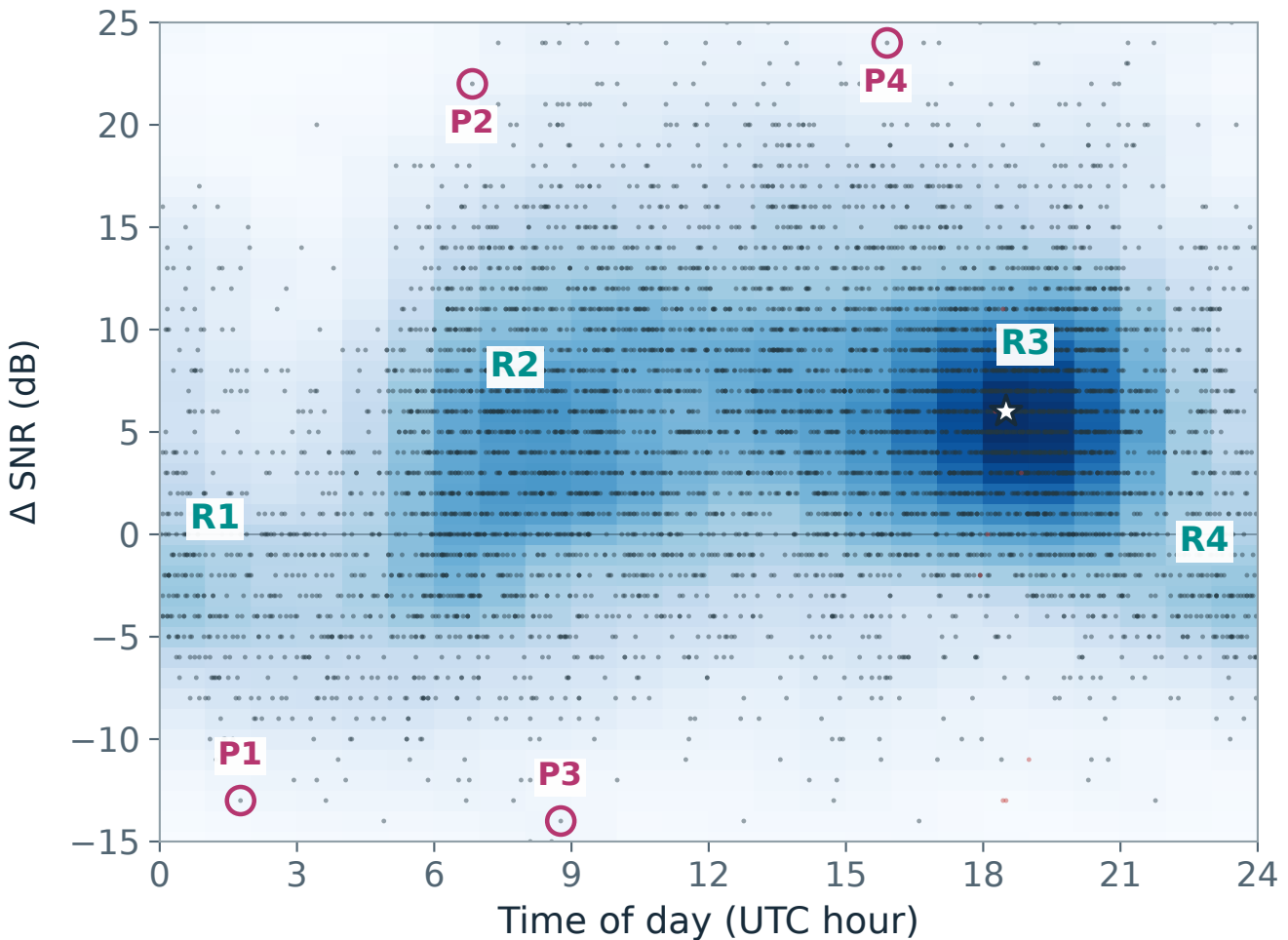
- R** R1-R4: paper regions (A-C)
- P** P1-P4: verification examples only (A-C)

Panel A: Original publication image, calibrated using its printed axes. R1-R4 locate the paper's regions.

P1-P4: Verification examples only, checking selected time-of-day and ΔSNR matches.

B Reconstruction

Smoothed observation density · linear dB



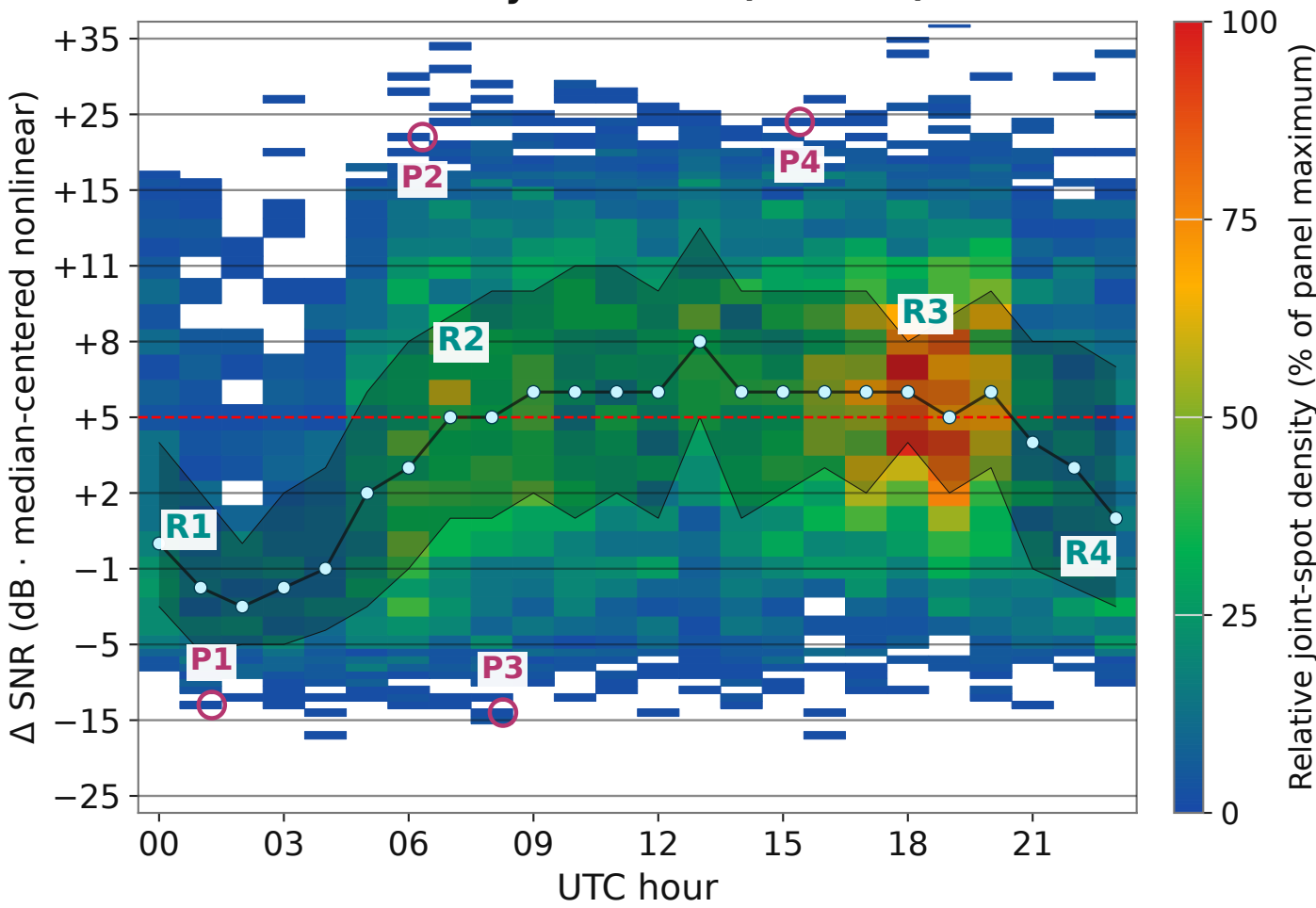
- Density / maximum (B)
- 0.00 0.10 0.25 0.50 1.00
- Individual paired observations (B)
 - Additional weaker-report combinations (B)
 - ☆ Highest reconstructed density (B)

Panel B: Paired observations recalculated through WSPRadar. Dark dots show retained pairs; red dots show additional* weaker-report combinations, excluded from density. WSPRadar retains the strongest report from each receiver per cycle/path. Shading shows the retained pairs' density with fixed smoothing.

*see Figure 3 **Pairing and duplicate reports**

C WSPRadar view

ΔSNR by UTC Hour (1 h bins)



- Median for each UTC hour (C)
- Middle 50% for each UTC hour (C)

Panel C: The same retained pairs in unsmoothed 1-hour × 1-dB cells, with hourly medians and the middle 50% (IQR). Read the native nonlinear dB axis carefully.